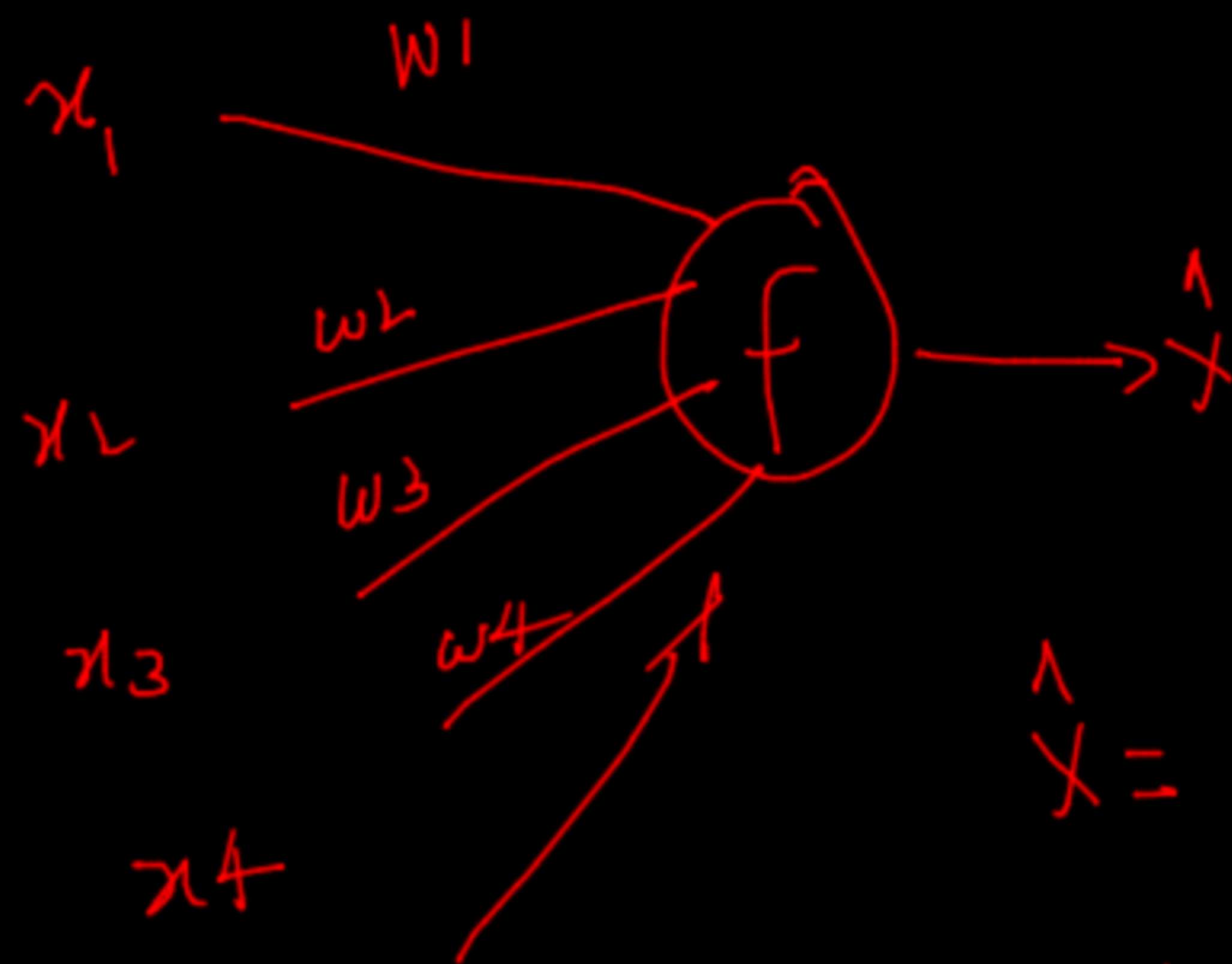


Draw a Single Neuron model

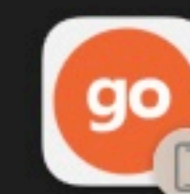


$$\hat{x} = W^T \cdot x + b$$

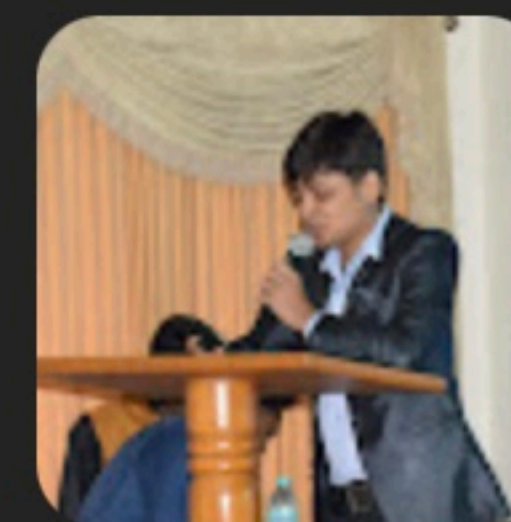
$$\hat{x} = \sum_{i=1}^n \omega_j \cdot x_{ij}$$

Linear Regression

$$\min \sum_{i=1}^n (y_i - \hat{y}_i)^2 + reg$$



Deal on Hotel
Use
for



Rahul Tripathi

→ Step 1 Define loss function (code)

nn (MSE loss)

$$d = \sum_{i=1}^n (y_i - \hat{y}) + \text{reg}(\text{all})$$

$$d_i = (y - \hat{y}_i)^2 \quad (\text{single points})$$

1) Identity → Linear Regression

2) Sigmoid → Log Regression

3) Threshold → perceptron

$w^T x$

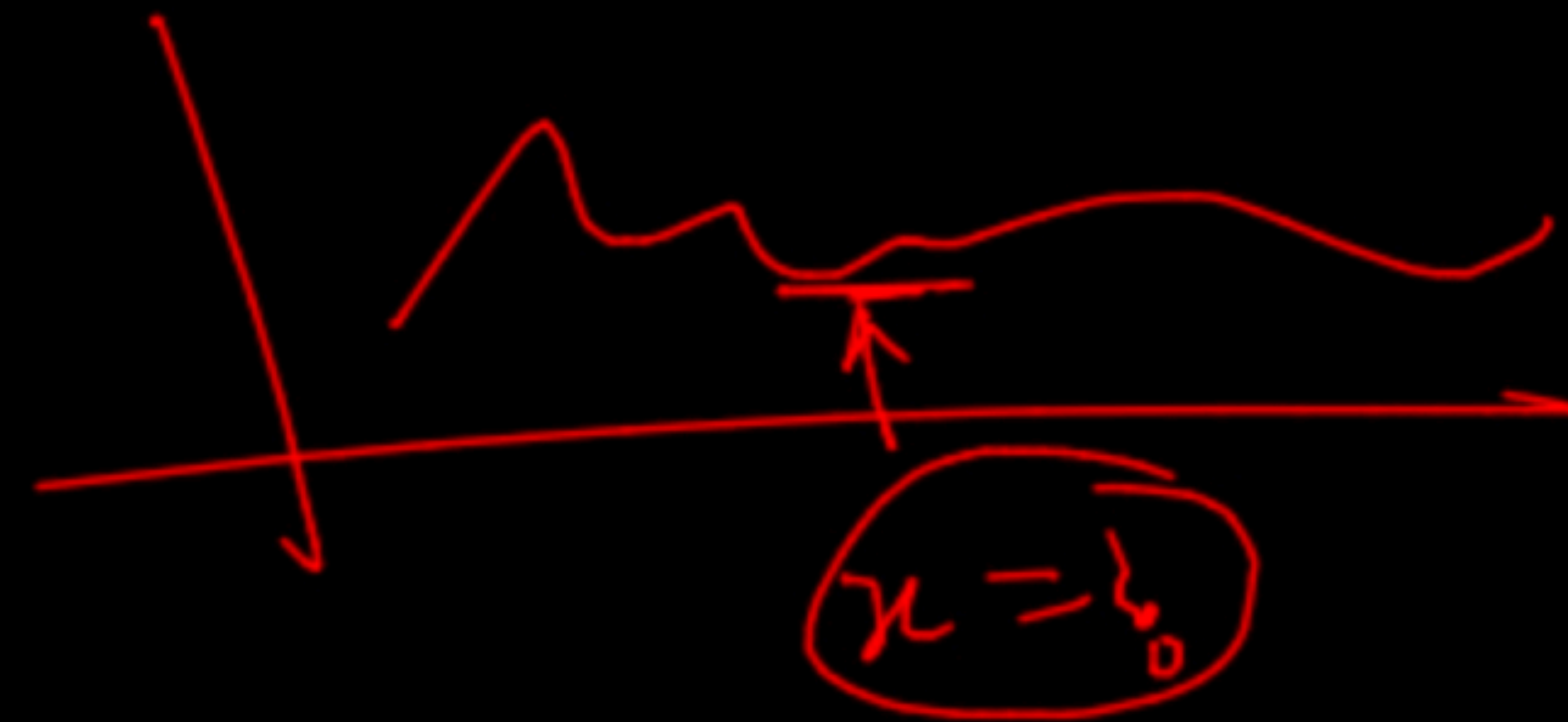
→ Step 2 Optimisation

$$\min_{w_i} \sum_{i=1}^n (y_i - \underbrace{w^T x_i}_{f(w^T x_i)})^2$$

→ Identity

math

$$\left\{ \begin{array}{l} y = f(x) \\ x \in [-1, 1] \\ y = f(x) \\ \downarrow \\ \text{mathe} \rightarrow \text{dot} \end{array} \right\}$$



Step 3: solve the optimisation problem using $\frac{\partial L}{\partial \omega}$

(3.1) Initialization of ω_i (random initialization)

(3.2) compute $\nabla_{\omega} L = \left[\frac{\partial L}{\partial \omega_1}, \frac{\partial L}{\partial \omega_2}, \frac{\partial L}{\partial \omega_3}, \frac{\partial L}{\partial \omega_4} \right]$

Talking: Rahul Tripathi



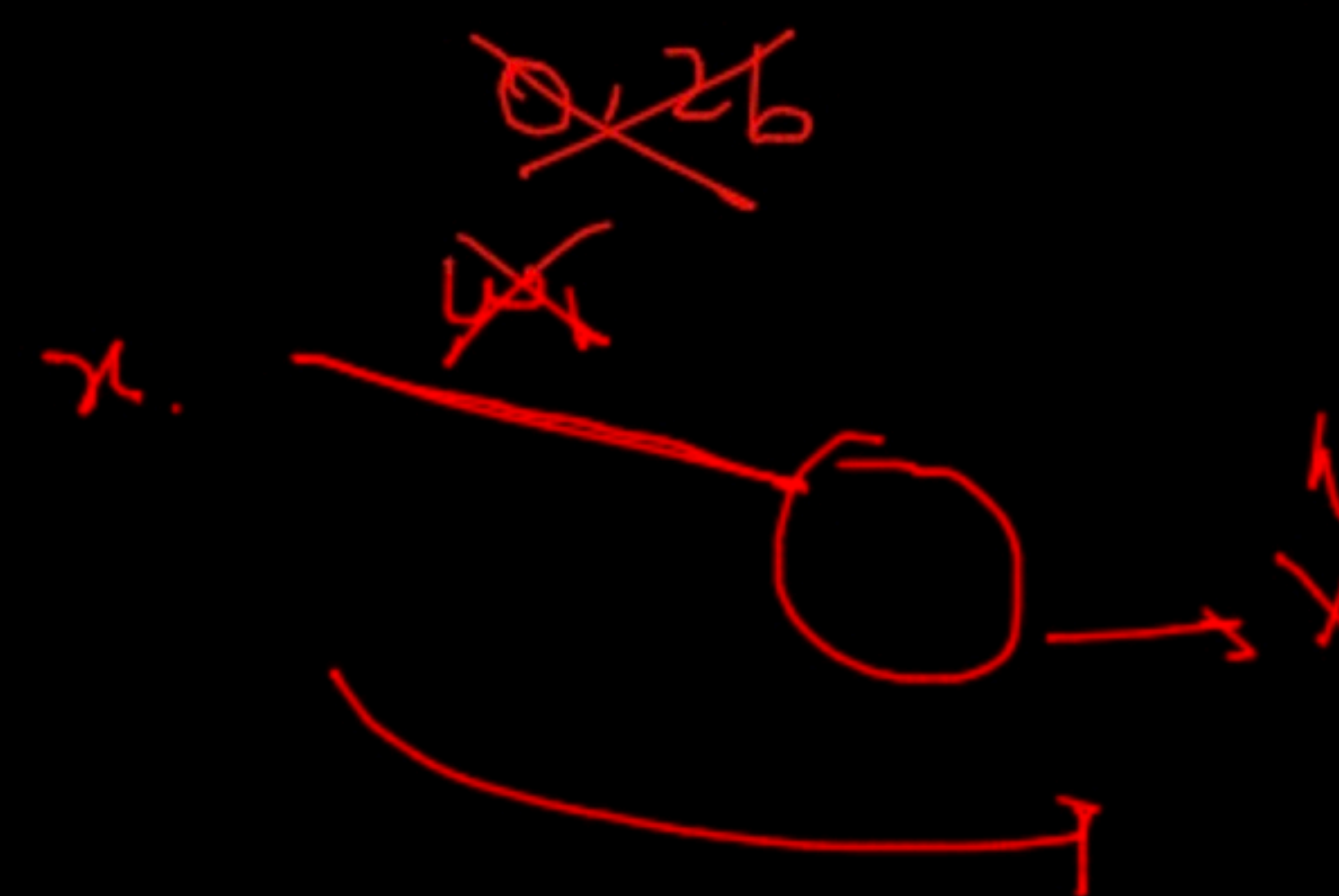
$$(w_i)_{new} = (w_i)_{old} - \eta \left[\nabla_{w_i} L \right]_{(w_{old})}$$

$$= 0.45 - 0.01 \times 0.26$$

$$(w_i)_{new} \approx \underline{\underline{0.45}} \text{ (approx)}$$

$$(w_i)_{old} = \underline{\underline{0.45}}$$

$$\left(\frac{\partial L}{\partial w_i} \right)_{w_i=1} = 0.26$$



0.45	0.45
0.26	0.26

$$(w_{new}) \approx (w_{old})$$

same $\rightarrow$ stop $\rightarrow$

Talking: Manshi Desai

$$\frac{\partial L}{\partial \omega_1} = \frac{\partial L}{\partial f} * \frac{\partial f}{\partial \omega_1}$$

$$d = (y - f(\omega^T x_i))^2$$

$$\frac{\partial L}{\partial \omega_1} = 2 (y_i - f(\omega^T x_i)) \frac{\partial (y - f(\omega^T x_i))}{\partial \omega_1}$$

$$\frac{\partial L}{\partial \omega_1} = -2x_i (y_i - f(\omega^T x_i))$$

Talking: Rahul Tripathi

← chain rule

$$\frac{\partial L}{\partial \omega_1} = \frac{\partial L}{\partial f} * \frac{\partial f}{\partial \omega_1}$$

code
↑

14

$$\frac{\partial L}{\partial f} = \sum_{i=1}^n 2(y - f(\bar{w}^T x_i))$$

↓

Talking: Rahul Tripathi

MLP (training)

(ANN)

→ RNN

↑ '2017'

↑ 'telugu' (image)

↑ 'fix'

(ANN)

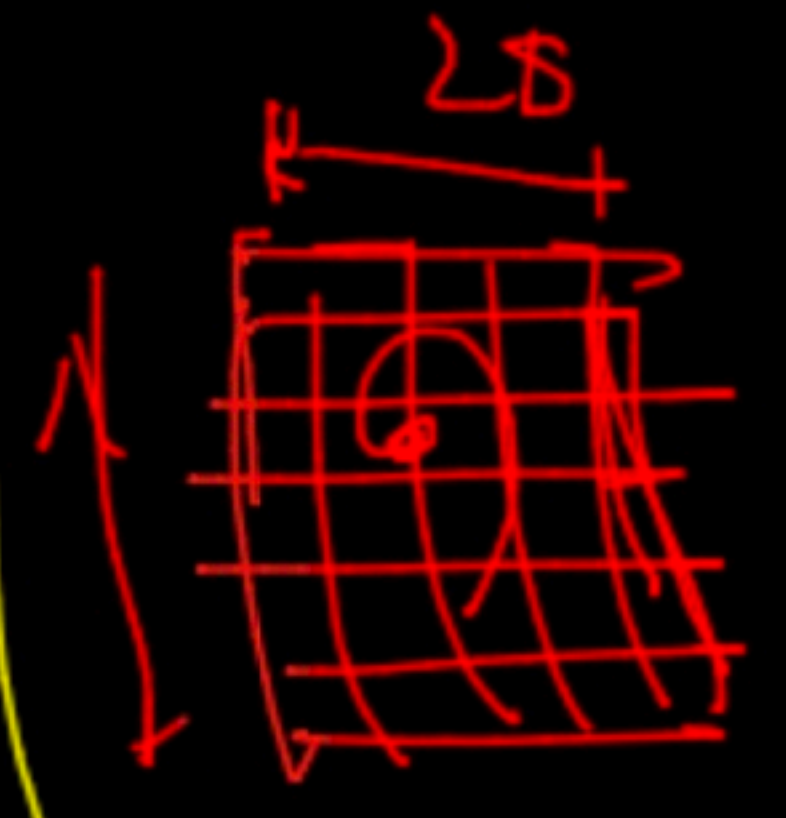
ANN

LSTM (HARD)

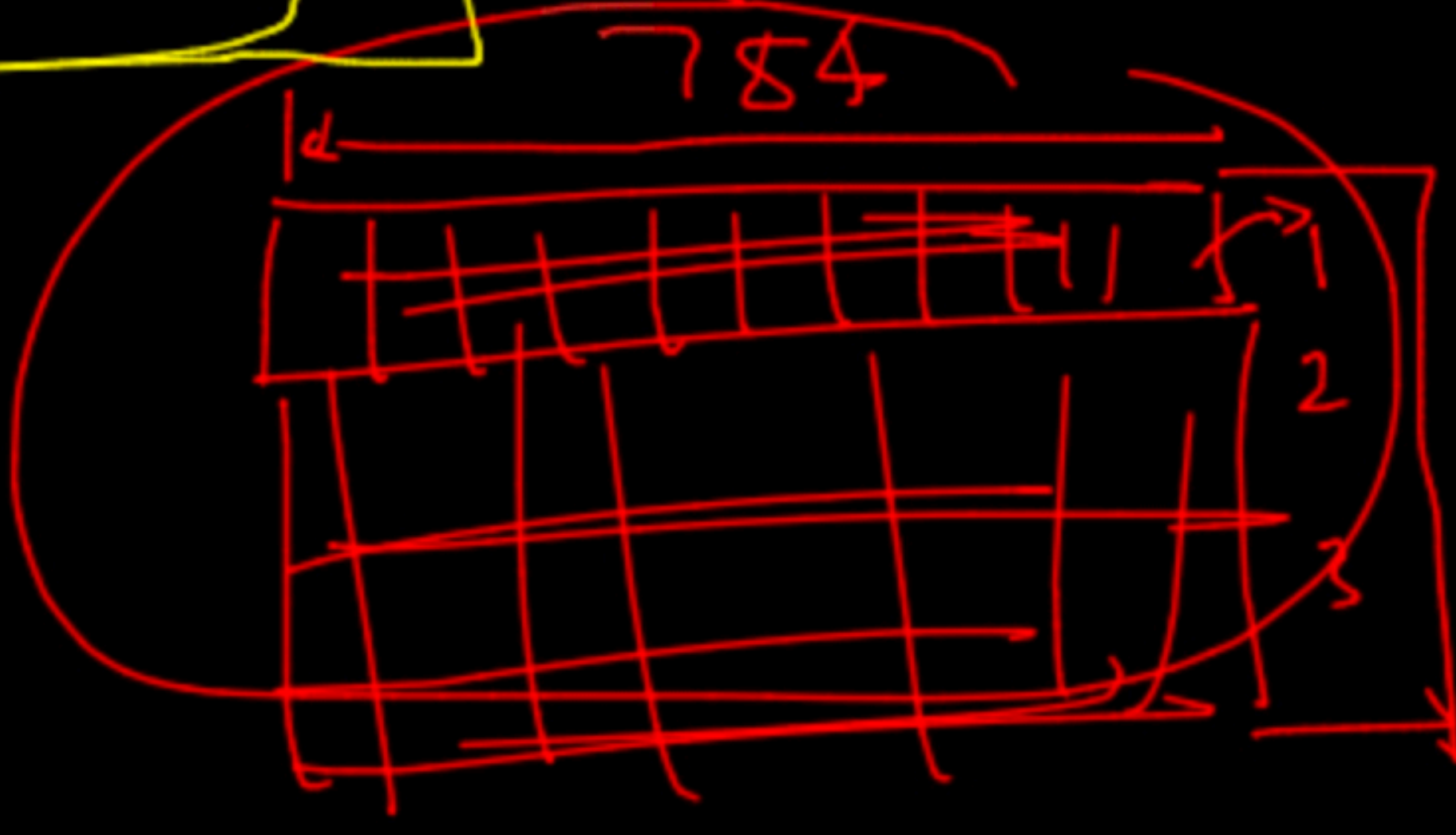
'seq - to - seq'

(text)

python

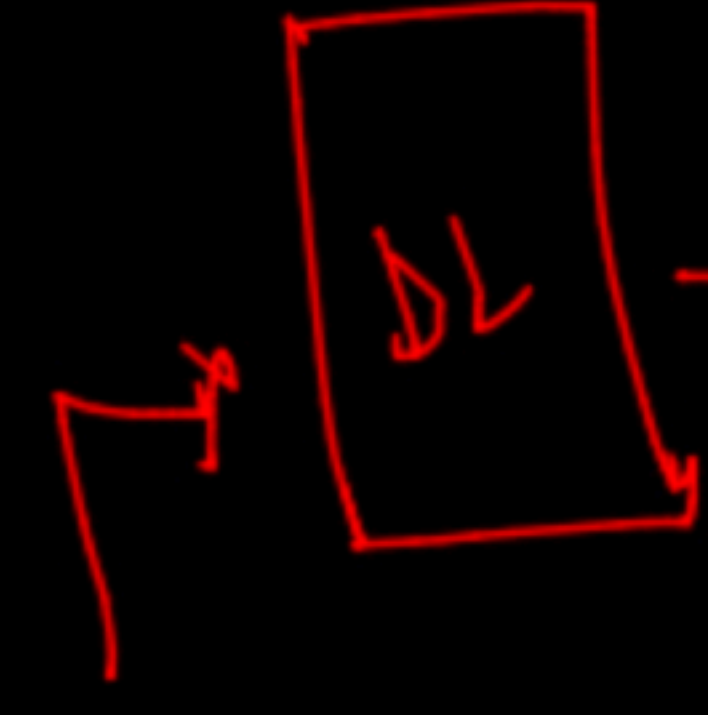


784



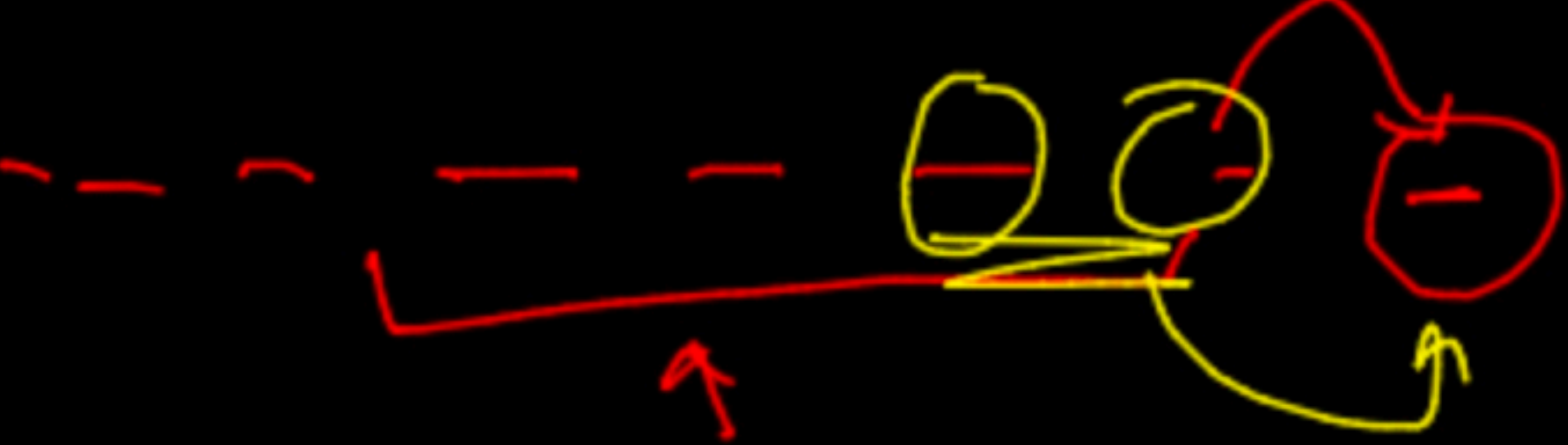
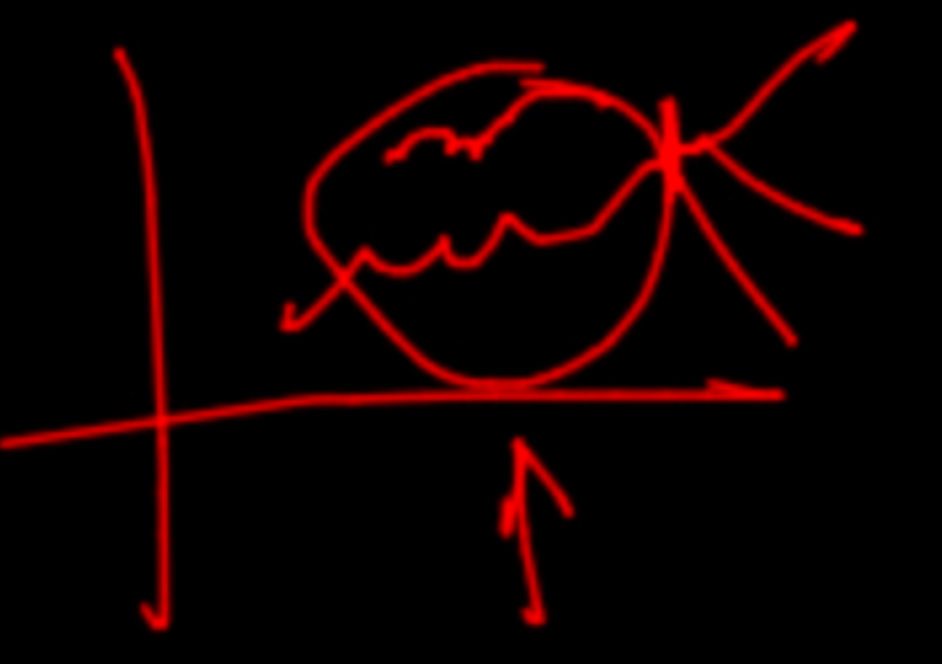
0
(Data)
(x,y)

input



'text'
(10/12)

'Forecasting'



Talking: Rahul Tripathi



Talking: Rahul Tripathi



9 / 9

you are screen sharing

Stop share